



ChipSat V6 (2023-2026)

Compact ChipSat with solar battery charging

Designed for suborbital rocket deployment and atmospheric data collection (DeSCENT mission)

Applications

- Low power, long range communications
- Remote sensing / distributed sensing
- Short-duration multi-day LEO operation
- Atmospheric or ground-based sensing

Dimensions: 60mm x 50mm x 4.5mm | **Mass:** 20g

Capabilities Summary

Power: Wireless self-sustaining operation via dual 300mW PV (one per side) + 300mAh LiPo battery

Processing: 32-bit ARM Cortex-M4; 48MHz; 256KB Flash, 65KB SRAM; 2MB external flash memory

Comms: Semtech SX126x; +22dBm TX; 915MHz ISM Band; LoRa, (G)FSK, (G)MSK, and BPSK

Position/Attitude determination: GNSS Receiver + 9-DOF IMU (accelerometer, gyroscope, magnetometer)

Environmental Sensing: Temperature, pressure, and humidity sensors; 3-DOF magnetometer from IMU

Survivability

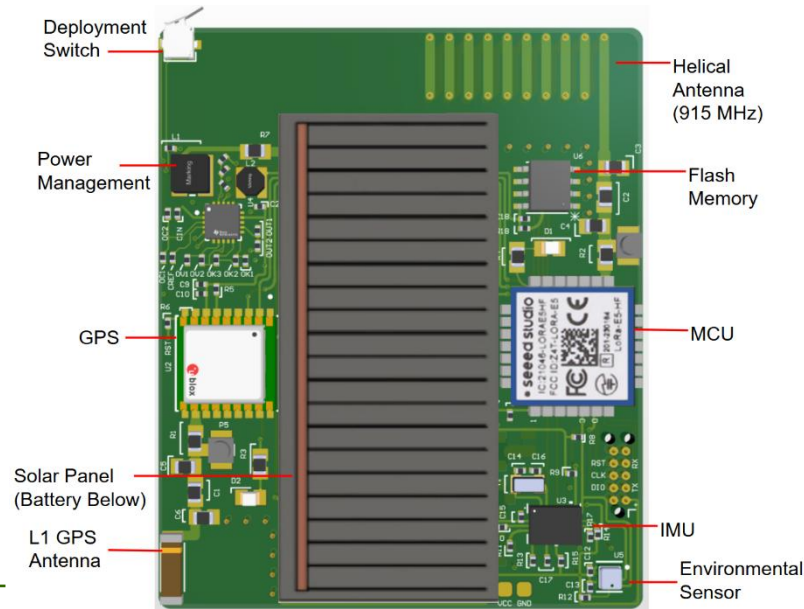
Vacuum ☒ | Launch loads/vibration ☒ | Radiation* ☒ | Terminal-velocity ground impact ☒

Operating temperature: -10°C – +60°C with LiPo battery | -40°C – +85°C with solar-only configuration

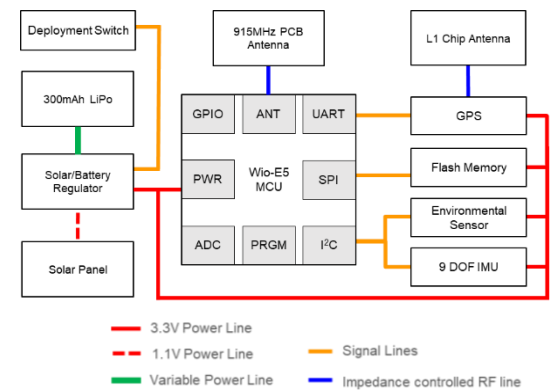
*No radiation shielding / rad-hard avionics –recommended for short-term missions w/ large quantities of ChipSats

Additional Specifications

MCU		Seed Studio Wio-E5 (STM32WLE5 32-bit ARM Cortex M4) + W25Q16 Ext Flash					
I/O Pins:		UART x3, I2C x1, SPI x1, 12-bit ADC x1, GPIO x6				Sleep current:	2.1 μ A
Clock Speed:		48MHz	Flash Memory:	256KB	SRAM:	65KB	Ext. Flash: 2MB
Sensors		BME280 Environmental			BNO085 9-DOF IMU		
		Temp	Pressure	Humidity	Accel	Gyro	Mag
Accuracy:		1°C	1.0 hPa	3% RH	0.3m/s ²	3.1 dps	1.4 μ T
Range:		-40°-85°C	300-1100 hPa	0-100%	± 8g	± 2000 dps	±800 μ T
GPS		u-blox MAX M10S; L1 GPS/Galileo/GLONASS/BeiDou; NOTE: CoCom Limits					
Accuracy:		2m	TTF:	2 min	Max alt:	80km	Max vel: 500m/s
Power		BQ25570 regulator IC + 300mAh LiPo Batt + 300mW GaAs PV					
Batt. Life:		10hr battery-only operation w/ frequent radio TX				Indefinite w/ solar charging	
Comms		SX126x (built into Wio-E5 MCU); also capable of (G)FSK, (G)MSK, and BPSK					
Freq:	915MHz	BW:	62.5 - 500KHz	TX PWR:	+22dBm	Modulation:	LoRa
Approx. Range:		(w/ below receiver equipment)		50km	LEO	GEO	Cislunar
Estimated LoRa Data rate (bps):		Low-gain omni		488 bps	91 bps	X	X
		High-gain Yagi		3.5 kbps	268 bps	36 bps*	X
		Large parabolic dish		22 kbps	1.9 kbps	91 bps	11 bps*
*requires hardware change							



ChipSat layout



ChipSat architecture

Est. cost:
\$200 USD
per ChipSat

For more information:
✉ mp336@cornell.edu
🌐 spacecraftresearch.com